

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks:20

Annexure A

Comparative Analysis

Code of Conduct:

I P Sai Bhavyatha(192424333)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Palacharla Sai Bhavyatha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies	
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive	
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided	
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated	
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand	

Comparative Analysis

1. CNNs vs. Vision Transformers (ViTs) for Object Detection

Scenario: A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time using limited labeled datasets.

Questions:

1. Compare CNNs and Vision Transformers in terms of:
 - Feature extraction capabilities for local vs. global context
 - Performance with small or sparse objects
 - Computational complexity and inference speed for embedded systems
2. Discuss which method is better suited for real-time deployment and how transfer learning or data augmentation could impact performance.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

Questions:

1. Compare PCA-based shape priors and graph-based learning in terms of:
 - Ability to handle non-rigid deformations
 - Robustness to varying mesh resolutions and noise
 - Ease of implementation and computational cost
2. Analyze which approach would provide more accurate correspondences for realistic morphing and why.

ANSWERS

1. CNNs vs. Vision Transformers (ViTs) for Object Detection

Scenario

A self-driving car company needs to detect rare urban obstacles such as traffic cones and animals in real time. The main challenges are limited labeled training data, small or sparsely located objects, high-resolution road images, and the computational limitations of embedded systems.

Question 1: Comparison of CNNs and Vision Transformers

A. Feature Extraction: Local vs. Global Context

CNNs (Convolutional Neural Networks):

CNNs use convolutional filters to extract spatial features from images. The initial layers learn simple features such as edges, corners, and textures, while deeper layers learn complex object patterns.

CNNs are particularly strong at **local feature extraction**. This makes them effective for recognizing the shape, texture, and boundaries of objects such as traffic cones and pedestrians.

However, conventional CNNs may require deeper layers or specialized architectures to capture long-range relationships between distant regions of an image.

Vision Transformers (ViTs):

Vision Transformers divide an image into patches and use self-attention to model relationships between different patches. Therefore, a ViT can consider both local and distant regions of an image.

For example, an obstacle can be interpreted together with its surrounding road, vehicle, pedestrian, and environmental context.

Comparison

Parameter	CNN	Vision Transformer
Basic operation	Convolution	Self-attention
Main strength	Local feature extraction	Global contextual relationships
Spatial features	Very strong	Strong
Global context	More limited in basic CNNs	Strong
Small object detection	Effective with multi-scale architectures	Can be effective with appropriate architecture
Training data requirement	Generally lower	Often benefits from large-scale pretraining
Embedded deployment	Easier	More computationally demanding

B. Performance with Small or Sparse Objects

Small objects such as traffic cones or distant animals may occupy only a small number of pixels in a high-resolution image.

CNN-based object detectors can use **multi-scale feature extraction** and feature pyramids to detect objects at different resolutions. This makes CNNs highly suitable for small-object detection.

Vision Transformers can capture relationships between an object and its surrounding scene using attention. This can help distinguish a small obstacle from visually similar background regions. However, the effectiveness depends strongly on the Transformer architecture, training strategy, and amount of available data.

Analysis

For a limited dataset, a well-designed CNN-based detector is generally easier to train effectively. A Transformer can become highly competitive when it benefits from suitable pretraining and sufficient computational resources.

C. Computational Complexity and Inference Speed

CNNs perform convolution operations that can be efficiently optimized using GPUs, NPUs, and other embedded accelerators. Lightweight CNN architectures can therefore achieve high inference speed with relatively low memory requirements.

Vision Transformers use self-attention mechanisms. For high-resolution images, attention can become computationally expensive, although modern architectures use techniques such as hierarchical attention, windowed attention, and efficient attention mechanisms to reduce this cost.

Factor	CNN	ViT
Computational requirement	Generally lower	Generally higher
Memory requirement	Lower for lightweight models	Can be higher
Inference speed	Usually faster	Can be slower
Embedded deployment	Highly suitable	Requires optimization
High-resolution processing	Efficient with suitable architecture	Can be computationally demanding
Hardware optimization	Mature	Increasingly supported

Question 2: Best Method for Real-Time Deployment

Answer

For the given autonomous-driving scenario, a **lightweight CNN-based object detector** is the most practical choice for real-time deployment on computationally constrained

embedded systems.

The reasons are:

1. CNNs provide efficient local feature extraction.
2. Lightweight CNN architectures can achieve low inference latency.
3. CNNs require comparatively fewer computational resources.
4. They work effectively with moderate-sized datasets.
5. Multi-scale CNN architectures are suitable for small-object detection.
6. CNN models are widely supported by embedded acceleration hardware.

However, a **CNN-Transformer hybrid** can also be considered if the vehicle hardware has sufficient computational capacity. The CNN can extract local features while the Transformer component captures global contextual information.

Role of Transfer Learning

Transfer learning can significantly improve performance when only a limited number of labeled obstacle images are available.

Instead of training a model from the beginning, a pretrained model is used as the starting point. The model has already learned general visual features such as edges, textures, shapes, and object patterns.

The process is:

Pretrained Model → Fine-Tuning with Road Dataset → Rare Obstacle Detection

This reduces training time and can improve performance when the target dataset is small.

Role of Data Augmentation

Data augmentation can artificially increase the diversity of training examples.

Useful augmentation techniques include:

- Rotation
- Scaling
- Cropping
- Brightness variation
- Contrast variation
- Horizontal flipping where appropriate
- Noise addition
- Partial occlusion
- Background variation

For rare obstacles, synthetic-data techniques such as GAN-based augmentation can also be considered.

Conclusion

CNNs are generally more suitable for real-time embedded deployment because of their computational efficiency, fast inference, and strong performance with limited data when combined with transfer learning and data augmentation. Vision Transformers provide excellent global-context modeling but may require greater computational resources. Therefore, an optimized CNN is the preferred baseline, while a hybrid CNN-Transformer approach can be considered for more powerful hardware.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence Scenario

An animation studio needs to align 3D human face meshes for realistic morphing applications. The input meshes may have different topology, vertex density, and geometric variations. The system must establish accurate correspondences between facial regions while handling non-rigid deformation and noise.

Question 1: Comparison of PCA-Based Shape Priors and Graph-Based Learning

A. Ability to Handle Non-Rigid Deformations

PCA-Based Shape Priors

PCA learns the main variations from a training collection of face shapes. A face can be represented using a mean shape and a set of principal components:

$$\mathbf{S} = \bar{\mathbf{S}} + \mathbf{P}\mathbf{b}$$

where:

- $\mathbf{S}$ = reconstructed face shape
- $\bar{\mathbf{S}}$ = mean face shape
- $\mathbf{P}$ = principal component matrix
- $\mathbf{b}$ = shape parameters

PCA works well when facial variations are relatively similar to the variations represented in the training data.

However, PCA mainly models variations as combinations of learned linear modes. Highly complex or non-linear deformations may not be represented accurately.

Graph-Based Learning

Graph-based learning represents the 3D face as a graph:

Vertices → **Nodes**

Mesh connections → **Edges**

The model learns relationships between neighboring vertices and larger structural patterns.

Graph-based methods can capture local geometric relationships and are more flexible for complex non-rigid shape variations.

Comparison

Parameter	PCA Shape Priors	Graph-Based Learning
Shape representation	Statistical model	Graph representation
Non-rigid deformation	Moderate	Strong
Complex deformation	Limited	Better suited
Training requirement	Relatively low	Usually higher
Flexibility	Lower	Higher

B. Robustness to Varying Mesh Resolutions and Noise

PCA-Based Shape Priors

PCA generally requires the training shapes to have compatible representations. If different meshes have different numbers of vertices or inconsistent topology, preprocessing and alignment are required.

Noise in the input can also affect the extracted shape representation. Appropriate smoothing and alignment can improve robustness.

Graph-Based Learning

Graph-based approaches naturally represent the relationships between vertices and their neighborhoods. They can therefore be more flexible when dealing with variations in mesh structure and vertex density, depending on the specific graph architecture.

Noise can be reduced using preprocessing, geometric filtering, or robust feature-learning methods.

Comparison

Parameter	PCA	Graph-Based Learning
Different vertex counts	Requires preprocessing	More flexible
Different topology	Difficult	Better suited
Noise handling	Moderate	Potentially stronger
Local geometric relationships	Limited	Strong
Mesh flexibility	Lower	Higher

C. Ease of Implementation and Computational Cost

PCA-Based Shape Priors

PCA is comparatively simple to implement and computationally efficient. The main steps include:

1. Align training shapes.
2. Convert shapes into vectors.
3. Calculate the mean shape.
4. Compute the covariance structure.
5. Extract principal components.
6. Reconstruct or compare shapes.

Because PCA is mathematically straightforward, it is easier to interpret and requires fewer computational resources.

Graph-Based Learning

Graph-based learning requires:

- Graph construction
- Feature extraction
- Model architecture
- Training
- Optimization
- Hyperparameter selection

It generally requires more computational resources and implementation effort.

Comparison

Parameter	PCA	Graph-Based Learning
Implementation difficulty	Low	High
Computational cost	Low–Moderate	Moderate–High
Interpretability	High	Moderate
Training complexity	Low	High
Scalability	Good for aligned shapes	Good with appropriate architecture

Question 2: Which Approach Provides More Accurate Correspondences?

Answer

For the given animation scenario, **graph-based learning is likely to provide more accurate correspondences for realistic 3D face morphing**, particularly when the input meshes have different topology, vertex density, and complex non-rigid deformations.

The main reasons are:

1. Better Geometric Representation

A graph preserves relationships between neighboring vertices. Facial structures such as the eyes, nose, mouth, and cheeks contain strong local geometric relationships that can be learned by graph-based models.

2. Handling Different Mesh Structures

Animation datasets may contain meshes with different numbers of vertices and different resolutions. Graph-based methods are more flexible for representing such structures than a direct PCA vector representation.

3. Non-Rigid Deformation

Human faces undergo complex changes when expressions such as smiling, speaking, or frowning occur. Graph-based learning can learn nonlinear relationships between facial regions and therefore handle such changes more effectively.

4. Local and Global Information

Graph-based models can learn both local neighborhood relationships and larger structural patterns. This is important for establishing accurate facial correspondences.

5. Better Morphing Quality

Accurate correspondence is essential for realistic morphing. Incorrect correspondence can cause artifacts such as distorted eyes, stretched mouths, or unnatural facial transitions.

Critical Evaluation

Criterion	PCA-Based Shape Priors	Graph-Based Learning
Simplicity	Excellent	Moderate
Computational cost	Low	Higher
Interpretability	Excellent	Moderate
Non-rigid deformation	Moderate	Excellent
Different mesh topology	Limited	Better
Different vertex	Requires	More flexible

Criterion	PCA-Based Shape Priors	Graph-Based Learning
density	preprocessing	
Noise robustness	Moderate	Good with suitable training
Local geometry	Limited	Strong
Global geometry	Good statistical representation	Strong
Morphing correspondence	Good for standardized meshes	Better for varied meshes

Synthesis and Insight

The comparison shows that **PCA and graph-based learning are useful for different situations.**

PCA is preferable when:

- The face meshes are standardized.
- Vertex correspondence is already available.
- The number of training samples is limited.
- Computational resources are limited.
- A simple and interpretable model is required.

Graph-based learning is preferable when:

- Meshes have different topologies.
 - Vertex density varies significantly.
 - Non-rigid facial deformation is important.
 - Local geometric relationships are important.
 - High-quality correspondence is required for realistic morphing.
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Final Conclusion

For the **autonomous vehicle obstacle detection problem**, CNNs are generally the better choice for real-time embedded deployment because they provide efficient feature extraction and fast inference. Vision Transformers provide stronger global-context modeling but can require more computational resources. Transfer learning and data augmentation are particularly important when labeled rare-obstacle data is limited.

For the **3D face correspondence problem**, PCA-based shape priors provide a simple, computationally efficient, and interpretable solution for standardized face meshes.

However, graph-based learning is more flexible for varying mesh topology, vertex density, noise, and complex non-rigid facial deformation. Therefore, **graph-based learning is the stronger choice for high-quality realistic face morphing**, while PCA remains an excellent baseline for simpler and more standardized datasets.